
Student Performance Prediction System

Team ID:PY-PT41
Problem statement ID: 16

Name : Diksha Rawat (Team Leader)

Class Roll Number: 41

University Roll Number : 2028710

Section and Group: P (G2)

Name : Rudransh Garg

Class Roll Number: 62

University Roll Number : 2028827

Section and Group: P (G2)

Name : Shubhangani kotiyal

Class Roll Number: 67

University Roll Number : 2028855

Section and Group: P (G2)

TABLE OF CONTENTS

1. Problem Description	3
1.1.Dataset Description.....	3
1.2.Flow diagram.....	5
2. Modules for Project Implementation.....	5
2.1. Modules Used	5
2.2. Platform Used	6
3. Machine Learning model/ Searching Technique used	7
4. Evaluation metrics used	8
5. Results and Visualizations	9
6. Conclusion and Future Scope	15
7. References.....	15

1. Problem Description

With the rapid growth of online platforms, users frequently share their opinions and reviews about products and services. These reviews contain valuable information that can help businesses understand customer satisfaction and improve their offerings. However, manually analyzing large volumes of text data is time-consuming and inefficient.

There is a need for an automated system that can accurately classify user reviews into different sentiment categories such as positive, negative, and neutral. This helps in quick decision-making and enhances user experience.

The problem addressed in this project is to develop a machine learning-based sentiment analysis model that processes textual review data, cleans and transforms it into numerical form, and classifies it using algorithms such as Logistic Regression, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM). The system aims to achieve reliable performance in predicting sentiment based on user reviews and ratings.

1.1 Dataset Description

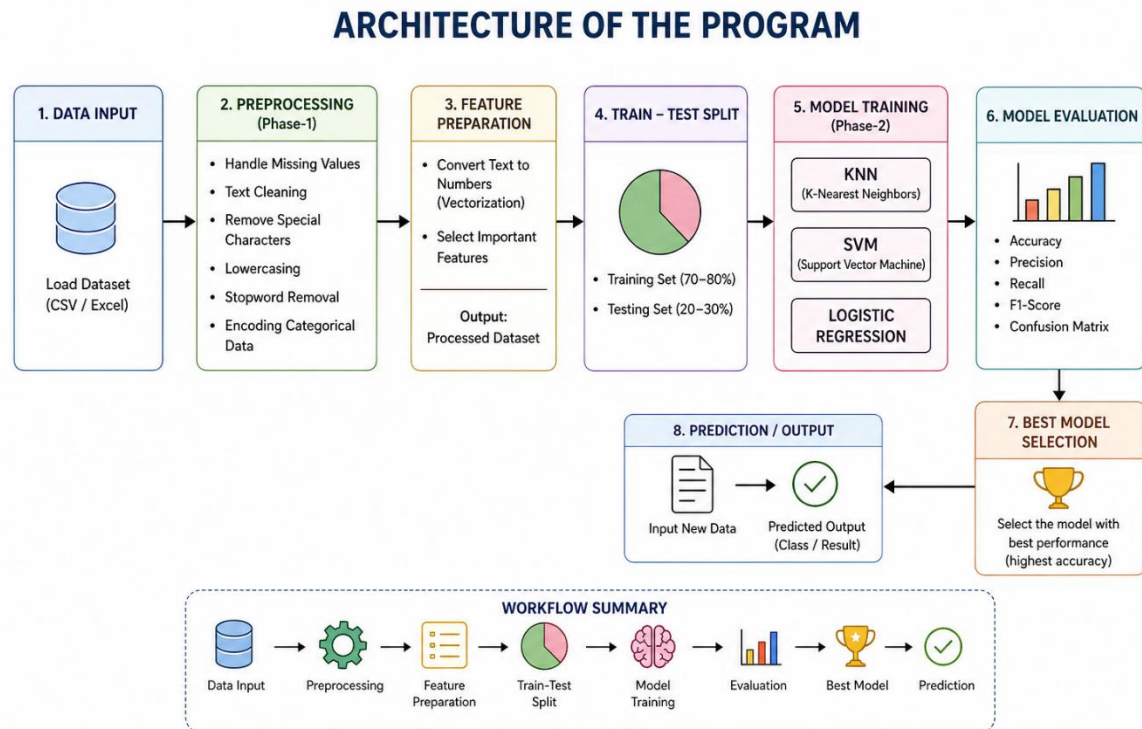
The dataset used in this project contains customer reviews of various products along with multiple attributes related to the product, user experience, and feedback. It is designed to analyze customer sentiment and predict whether a review is positive, negative, or neutral.

Features (Columns)

1. **Review_ID**
A unique identifier assigned to each review. It ensures that every record in the dataset can be uniquely distinguished.
2. **Product_Name**
The name of the product being reviewed.
3. **Category**
The category to which the product belongs (e.g., electronics, accessories, etc.).

4. Brand
The brand or manufacturer of the product.
5. Price_INR
The price of the product in Indian Rupees.
6. Rating
A numerical value (typically between 1 and 5) given by the user to rate the product.
7. Review_Title
A short title summarizing the user's review.
8. Review_Text
The main textual feedback provided by the user. This is the most important feature for sentiment analysis.
9. Review_Date
The date on which the review was posted.
10. Verified_Purchase
Indicates whether the user actually purchased the product (1 = Yes, 0 = No).
11. Helpful_Votes
The number of users who found the review helpful.
12. Location
The geographical location of the reviewer.
13. Device_Age_Months
The age of the product in months at the time of review.
14. Would_Recommend
Indicates whether the user would recommend the product (0 = No, 1 = Maybe, 2 = Yes).

1.2 Flow Diagram



2. Modules for Project Implementation

2.1 Modules Used

The following inbuilt and external Python modules were used in this project:

1. NumPy

- Used for numerical operations and handling arrays.
- Helps in combining feature vectors and ratings efficiently.

2. Pandas

- Used for data manipulation and analysis.
- Helps in reading datasets, cleaning data, and handling dataframes.

3. Scikit-learn (sklearn)

A major library used for machine learning tasks:

- **LogisticRegression**
Used for classification of sentiments.
- **KNeighborsClassifier (KNN)**
Used to classify data based on nearest neighbors.
- **SVC (Support Vector Machine)**
Used for advanced classification with better decision boundaries.
- **train_test_split**
Used to divide dataset into training and testing sets.
- **TfidfVectorizer**
Converts text data into numerical feature vectors.
- **StandardScaler**
Used to normalize/scale numerical data.
- **metrics (accuracy_score, confusion_matrix, classification_report)**
Used to evaluate model performance.

4. NLTK (Natural Language Toolkit)

- Used for text preprocessing.
- Includes:
 - Stopwords removal
 - Tokenization
 - Stemming

5. Re (Regular Expressions)

- Used for cleaning text data.
- Helps in removing punctuation, special characters, etc.

6. String Module

- Used for handling string operations like removing punctuation.

7. Pickle

- Used to save and load trained models.

2.2 Platform Used:

The development and implementation of this project were carried out using the following platforms:

1. Jupyter Notebook

- Used as the primary development environment.
- Helps in writing and executing code step-by-step.
- Useful for data visualization, testing models, and documenting results.

2. Python

- Programming language used for the entire project.
- Provides strong support for machine learning and data analysis.

3. Anaconda (Optional)

- **Used for managing Python environments and packages.**
- **Simplifies installation of required libraries.**

2. Machine Learning Model/ Searching Technique Used

Logistic Regression

- A linear classification algorithm.
- Used to predict the probability of a review belonging to a particular sentiment class.
- Works well for text classification problems with high-dimensional data.
- Provides fast and efficient results.

2. K-Nearest Neighbors (KNN)

- A non-parametric, instance-based learning algorithm.
- Classifies a review based on the majority class of its nearest neighbors.
- Uses distance metrics (like Euclidean distance) to find similar data points.
- Simple but computationally expensive for large datasets.

3. Support Vector Machine (SVM)

- A powerful classification algorithm.
- Finds the optimal hyperplane that separates different sentiment classes.
- Effective in high-dimensional spaces like text data.
- Provides high accuracy compared to other models.

Searching Technique Used

K-Nearest Neighbors Search

- The KNN algorithm uses a distance-based searching technique.
- It searches for the 'K' closest data points in the feature space.
- Based on these neighbors, it assigns the sentiment class.
- Common distance metric used:
 - Euclidean Distance

Feature Extraction Technique

TF-IDF (Term Frequency – Inverse Document Frequency)

- Converts text data into numerical form.
- Assigns importance to words based on their frequency in a document and across all documents.
- Helps improve model accuracy by reducing the impact of common words.

4. Evaluation metrics used (Describe each evaluation metric used along with formulae)

1. Accuracy

- Measures the overall correctness of the model.
- It is the ratio of correctly predicted observations to total observations.

Formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

2. Precision

- Measures how many predicted positive cases are actually positive.
- Important when false positives are costly.

Formula:

$$Precision = \frac{TP}{TP + FP}$$

3. Recall (Sensitivity)

- Measures how many actual positive cases are correctly identified.
- Important when missing positive cases is critical.

Formula:

$$Recall = \frac{TP}{TP + FN}$$

4. F1-Score

- Harmonic mean of Precision and Recall.
- Useful when there is class imbalance.

Formula:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

5. Confusion Matrix

- A table used to describe the performance of a classification model.
-

Actual Positive	TP	FN
Actual Negative	FP	TN

- Helps visualize model performance in detail.

6. Classification Report

- A summary of:
 - Precision
 - Recall
 - F1-score
 - Support (number of actual samples)
- Generated using sklearn:

from sklearn.metrics import classification_report

7. ROC Curve (Receiver Operating Characteristic)

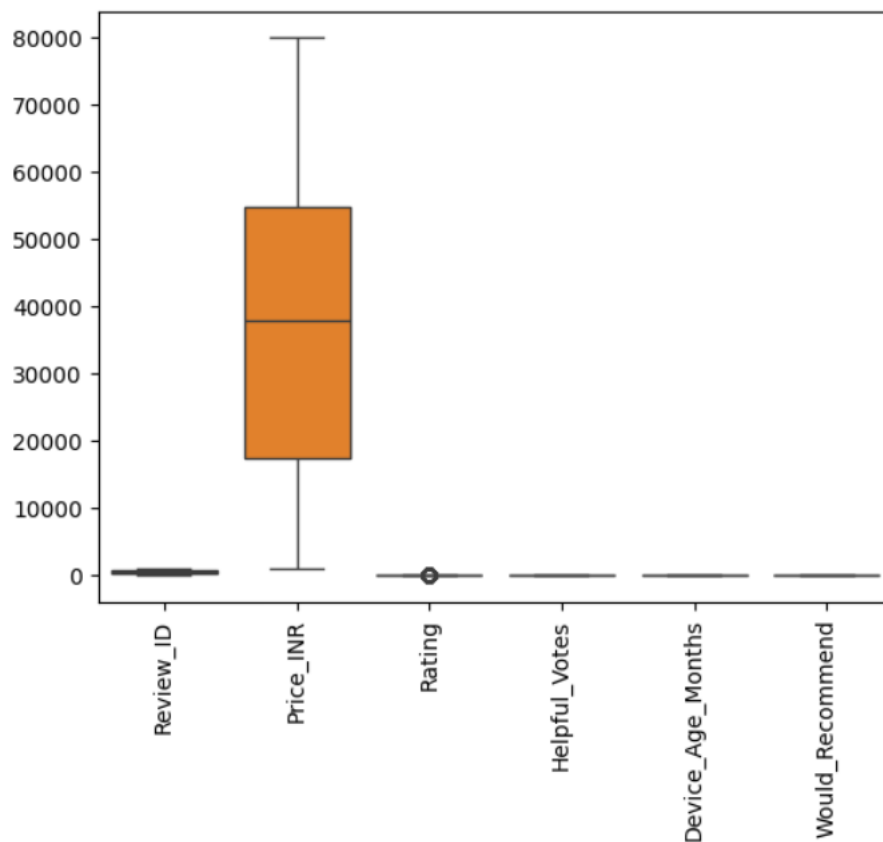
- A graph showing the performance of a classification model at different thresholds.
- Plots:
 - True Positive Rate (TPR) vs False Positive Rate (FPR)

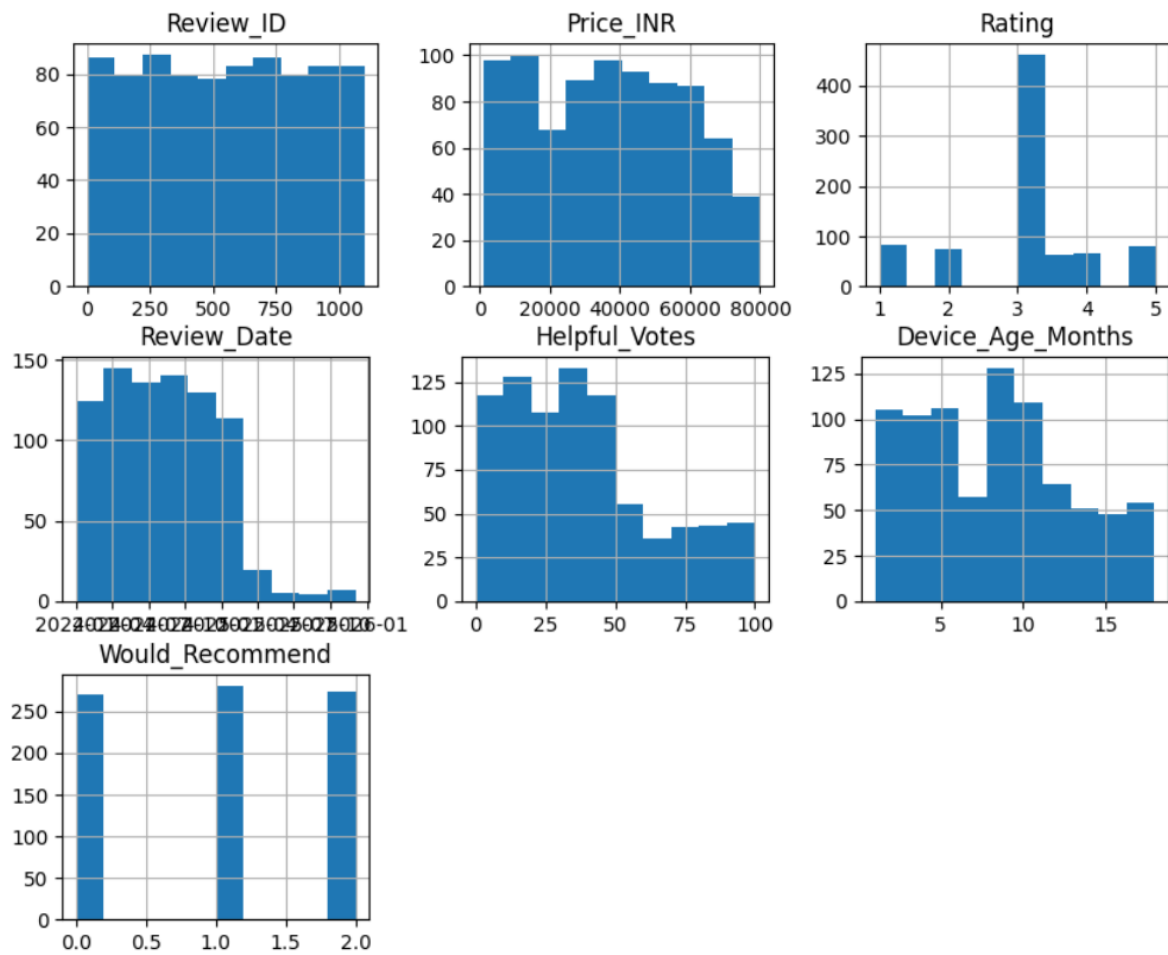
Formulas:

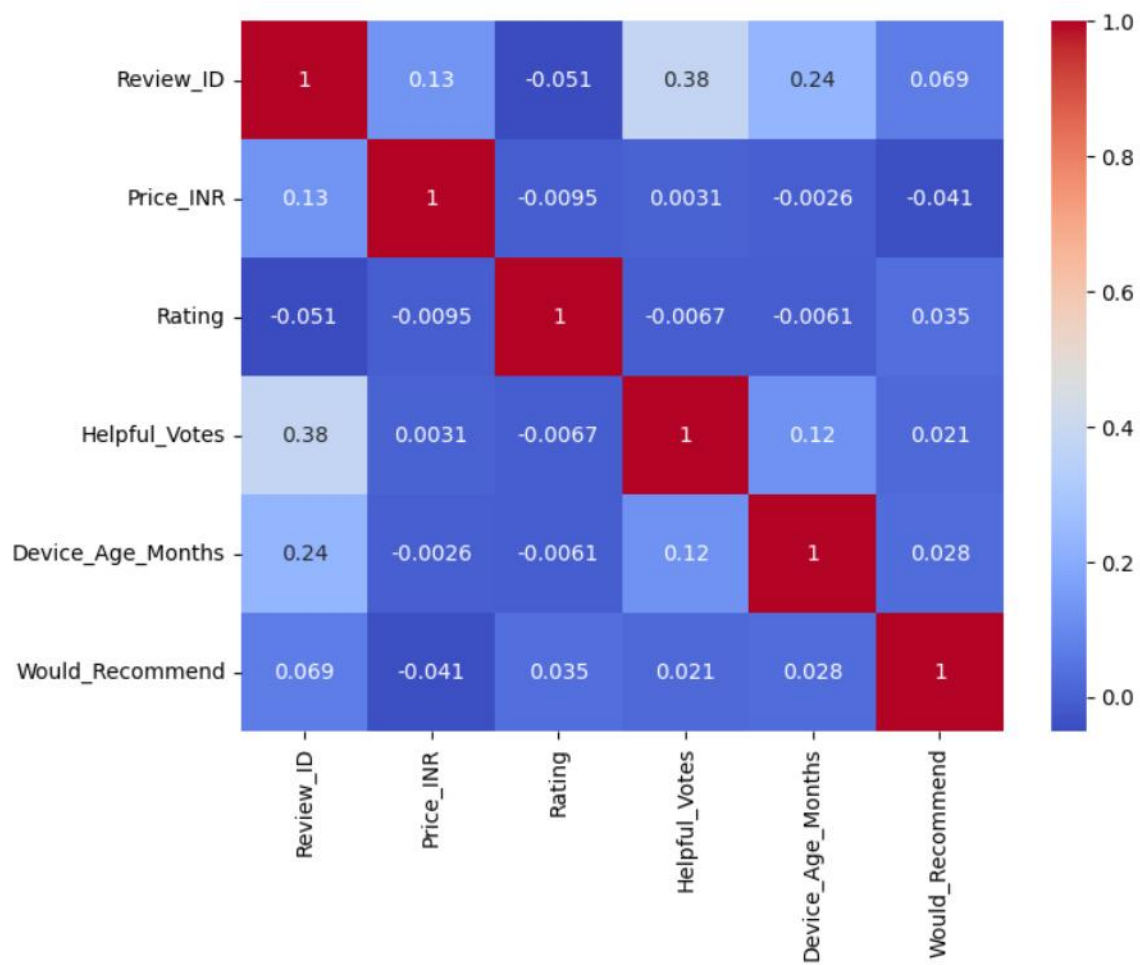
$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

5. Results and Visualizations







Logistic Regression

Accuracy: 0.8629032258064516

	precision	recall	f1-score	support
Negative	1.00	1.00	1.00	50
Neutral	0.00	0.00	0.00	34
Positive	0.83	1.00	0.91	164
accuracy			0.86	248
macro avg	0.61	0.67	0.64	248
weighted avg	0.75	0.86	0.80	248

KNN

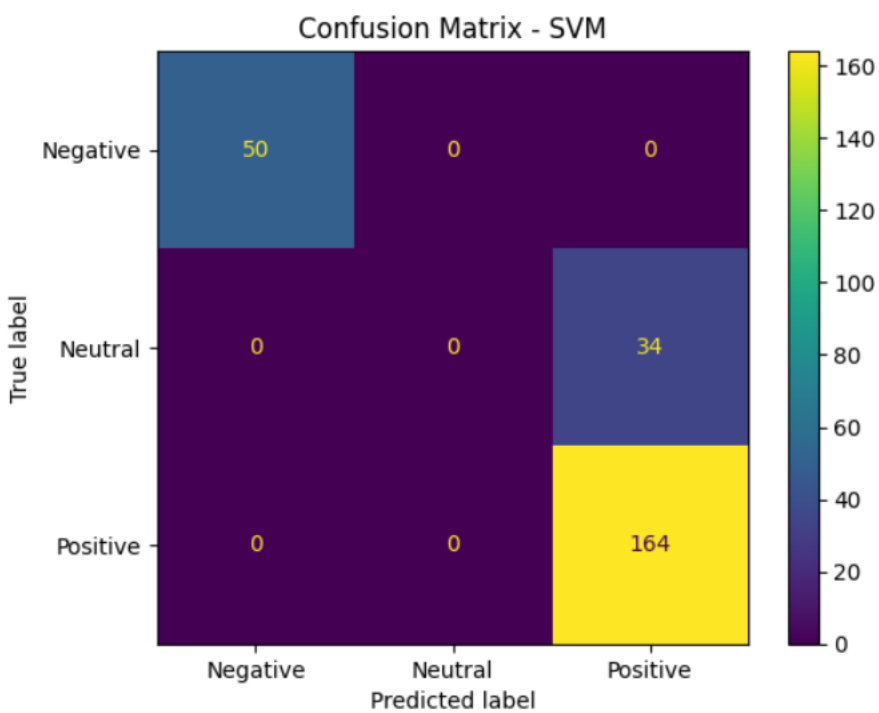
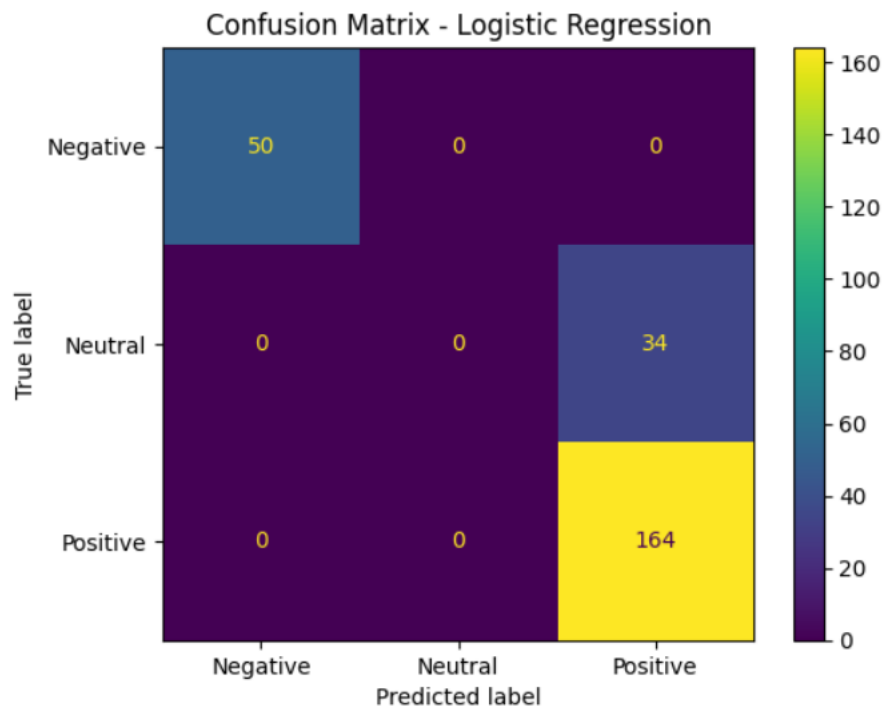
Accuracy: 0.9758064516129032

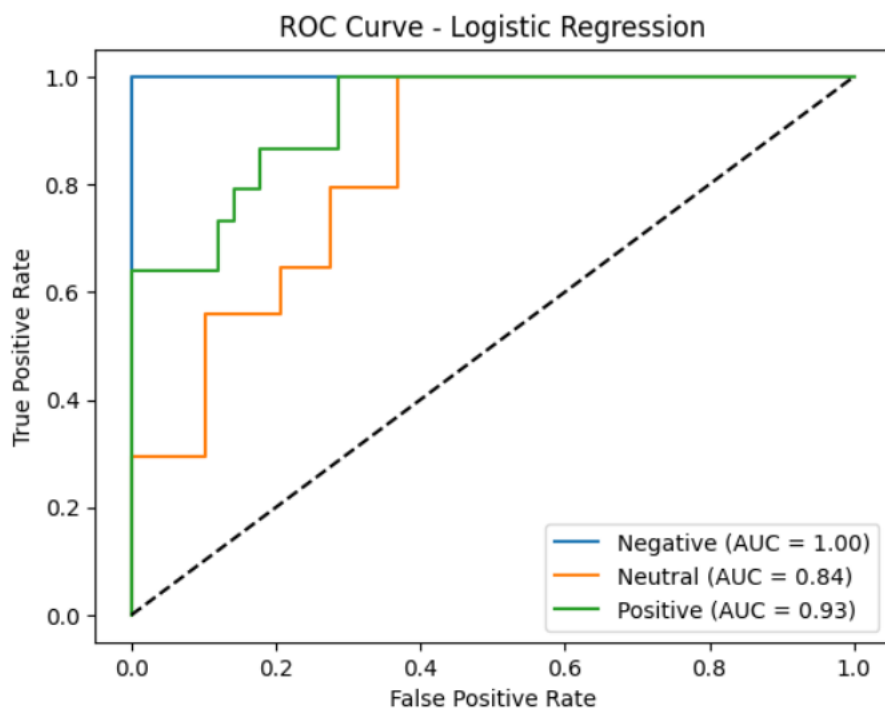
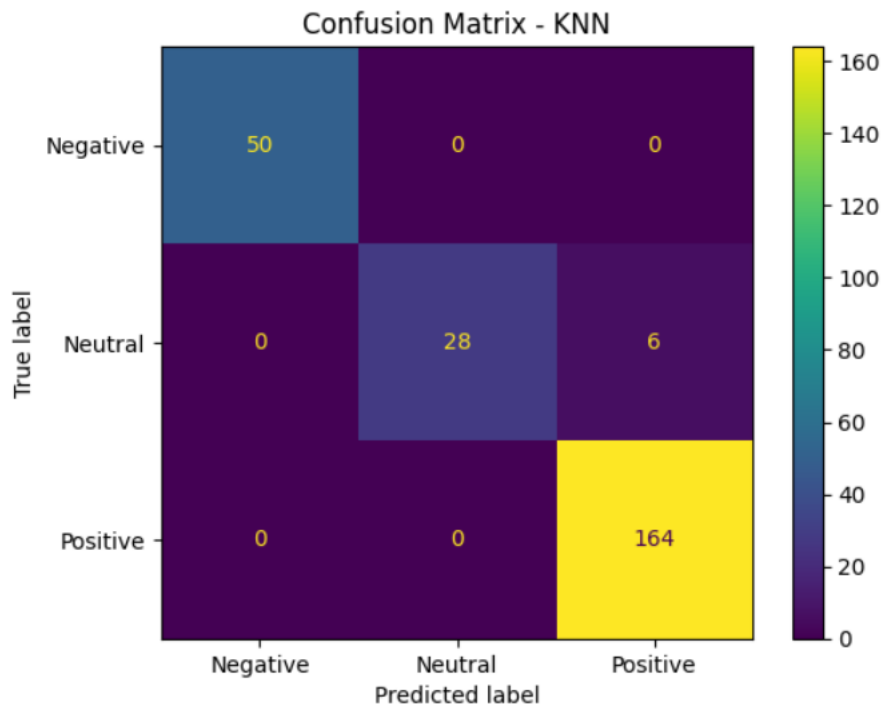
	precision	recall	f1-score	support
Negative	1.00	1.00	1.00	50
Neutral	1.00	0.82	0.90	34
Positive	0.96	1.00	0.98	164
accuracy			0.98	248
macro avg	0.99	0.94	0.96	248
weighted avg	0.98	0.98	0.97	248

SVM

Accuracy: 0.8629032258064516

	precision	recall	f1-score	support
Negative	1.00	1.00	1.00	50
Neutral	0.00	0.00	0.00	34
Positive	0.83	1.00	0.91	164
accuracy			0.86	248
macro avg	0.61	0.67	0.64	248
weighted avg	0.75	0.86	0.80	248





6. Conclusion and Future Scope:

In this project, a machine learning-based sentiment analysis system was successfully developed to classify user reviews into different sentiment categories. The system preprocesses raw text data, converts it into numerical form using TF-IDF, and applies multiple classification algorithms such as Logistic Regression, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM).

The models were evaluated using various performance metrics like accuracy, precision, recall, and F1-score. Among the implemented models, SVM and Logistic Regression provided better accuracy and performance compared to KNN for text classification tasks. Overall, the project demonstrates how machine learning can be effectively used to analyze large volumes of textual data and extract meaningful insights automatically, reducing manual effort and improving decision-making.

Future Scope

The project can be further enhanced in the following ways:

- **Use of Deep Learning Models:**
Implement advanced techniques like LSTM, RNN, or Transformers for better accuracy.
- **Real-Time Sentiment Analysis:**
Integrate the system with live data sources such as social media or websites.
- **Deployment as a Web Application:**
Convert the model into a user-friendly web app using frameworks like Flask or Django.
- **Multilingual Support:**
Extend the model to analyze reviews in multiple languages.
- **Improved Data Handling:**
Use larger and more diverse datasets for better generalization.
- **Aspect-Based Sentiment Analysis:**
Analyze sentiments for specific aspects (e.g., price, quality, service).

7. References:

Books:

1. Let Us Python – Yashavant Kanetkar

Online Resources:

- GeeksforGeeks – <https://www.geeksforgeeks.org/>
- TutorialsPoint – <https://www.tutorialspoint.com/cprogramming/index.htm>
- Stack Overflow – <https://stackoverflow.com>